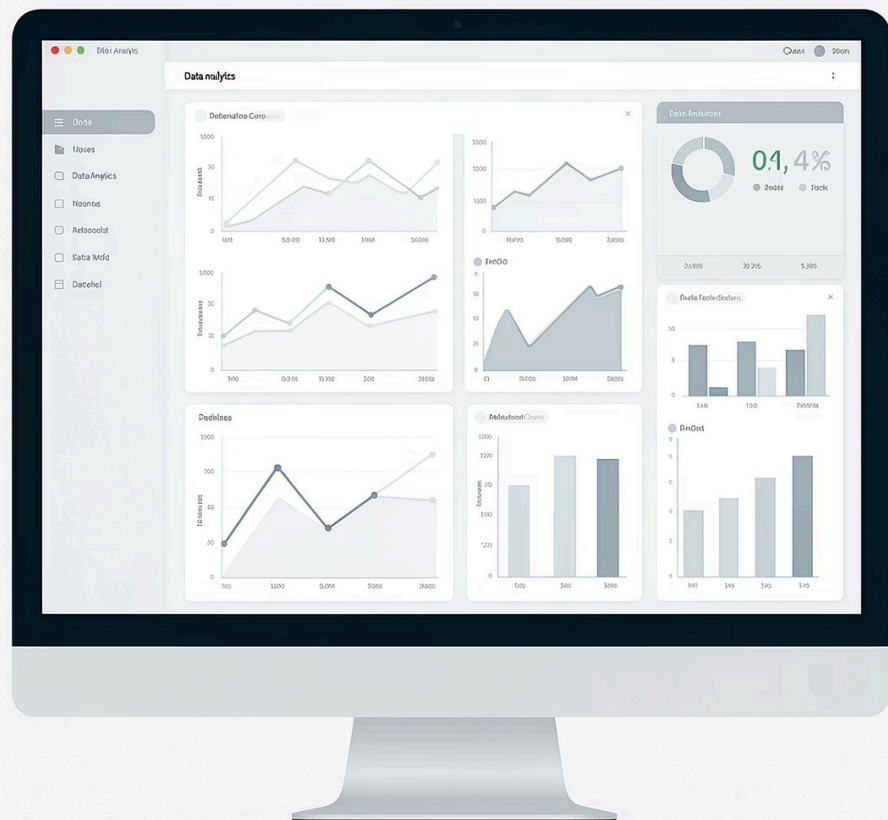




DVA CAPSTONE

Food & Hospitality Analytics | Group 15

Faculty Mentor: Archit Raj

Team Members: Nistha Gupta, Adil Mirza, Tanish Yadav, Tejas Tyagi, Kaustubh, Krishiv

GitHub: https://github.com/Kaustubh0505/C_G15_dvaCapstone | Tableau: https://public.tableau.com/app/profile/adil.mirza1032/viz/Book1_17772634695820/HOME

Context & Problem Statement

Sector Context

Restaurant businesses rely on ratings, pricing, and location insights to improve revenue and customer engagement. This matters for restaurant owners, aggregators, and business managers.

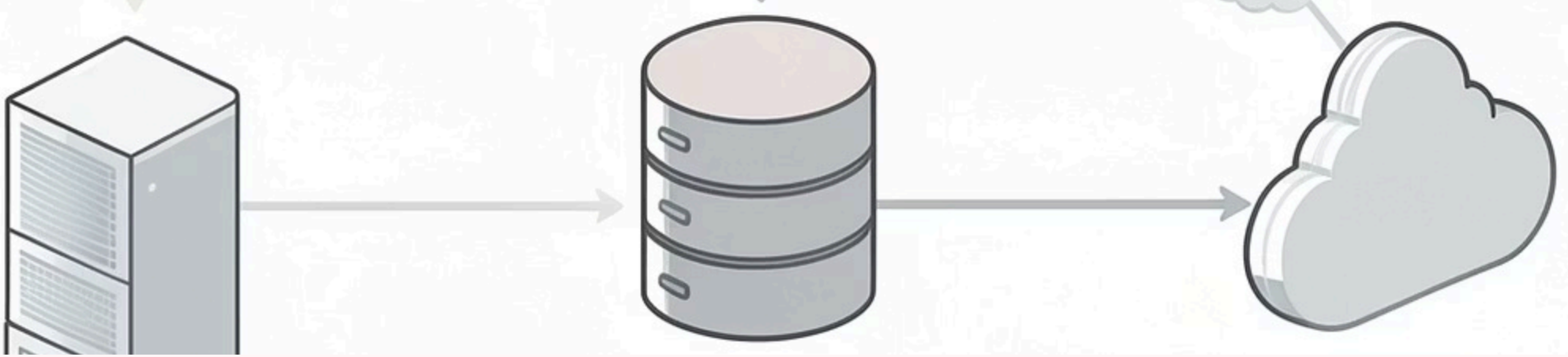
Problem Statement

Which types of restaurants (based on location, cuisine, and cost) achieve higher ratings and customer engagement (votes), and how does this vary across cities?

Objective

Support decisions on pricing, positioning, and customer targeting through data-driven insights.





Data Engineering: Source to Pipeline

Dataset Overview

- Source: [Zomato Bangalore Restaurants Dataset, (51,717 rows, 11 columns), time period]
- **Cleaning Steps:** Null Handling, Outlier Treatment, Standardization

GitHub: Notebooks: 01_data_cleaning.ipynb & 02_feature_engineering.ipynb

Data Dictionary

[**Name**]: Name of the restaurant

[**online_order**]: Indicates whether online ordering is available

[**book_table**]: Indicates whether table booking is available

[**rate**]: Average rating of the restaurant (out of 5)

KPI & Metrics Framework

What are we measuring?

3.70

Average Rating

Customer satisfaction.

₹554.4

Average Cost for Two

Pricing level.

14.67M

Total Votes

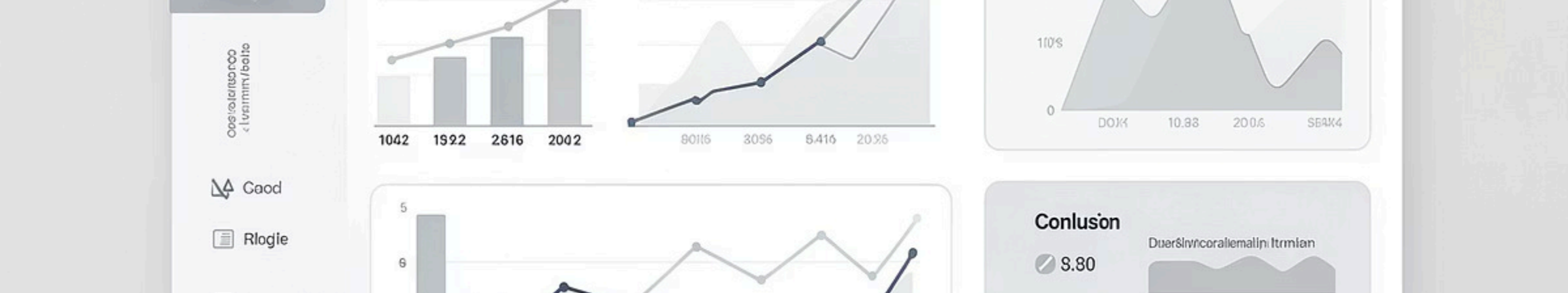
Popularity and engagement.

Why these KPIs?

- Evaluates service performance
- Supports pricing decisions
- Measures demand and popularity

Link to Business Problem:

Understand factors driving restaurant performance and customer preference. These KPIs directly support decisions on pricing, quality improvement, and market opportunity.



Key Insights (EDA)

Ratings Cluster Around 3.7

Most restaurants concentrate around stable mid-to-high satisfaction levels.

Popularity Is Highly Concentrated

A small set of restaurants captures disproportionate customer attention (right-skewed votes).

Premium Pricing ≠ Better Ratings

Higher-cost restaurants do not consistently achieve proportionately better ratings.

Online Ordering Shapes Pricing

Restaurants offering online orders show different pricing patterns, reflecting digital-first demand.

Moderate Positive Correlations

Ratings, votes, and pricing move together in meaningful but non-linear ways (rate-votes ~ 0.43 , book_table-cost ~ 0.62).

Source: Insights derived from 03_eda.ipynb

Statistical Testing, Prediction and Segmentation Revealed Deeper Market Structure

Hypothesis Testing

Methods: Mann-Whitney U Test, Chi-Square Test (scipy.stats)

Finding: Online ordering associated with significant rating differences; Table booking strongly relates to higher votes

Significance: p-values ≈ 0 (very strong)

Predictive Modeling

Method: Random Forest Regressor (sklearn)

Predicted: Restaurant ratings & customer votes

Performance: Rating $R^2 \approx 0.80$ | Votes $R^2 \approx 0.91$

Finding: Feature interactions explain and predict performance

Segmentation

Method: K-Means Clustering (StandardScaler)

Segments: Budget/Low-engagement | Mid-market | Premium/High-engagement

Finding: Restaurants naturally group into strategic market segments

Beyond descriptive EDA, advanced analysis validated relationships statistically, predicted performance drivers, and uncovered market segments.

Libraries: scipy | sklearn | pandas | numpy

Tableau Dashboard Walkthrough

Executive View

- Provides a management-level summary of restaurant ecosystem performance
- Tracks customer satisfaction, demand, and pricing at a glance
- Designed for quick strategic monitoring

Operational View

- Drill-down analysis reveals pricing vs ratings by cuisine
- Location-wise rating heatmap identifies premium cuisines, underserved locations, and pricing patterns

Interactive Filters

Interactive filters allow users to dynamically explore ratings and pricing by cuisine, location, and service attributes (online order, table booking).

Tableau Public URL: [View Dashboard on Tableau Public](#)

Recommendations

Promote mid-tier restaurants

Insight: Popularity is concentrated among a small set of restaurants

Recommendation: Invest in promoting mid-tier restaurants through visibility boosts, offers, and discovery algorithms

Expected Impact: More balanced customer traffic, reduced overdependence on a few popular outlets, higher platform-wide engagement

Expand digital ordering support

Insight: Online ordering is associated with stronger performance and rating differences

Recommendation: Expand digital ordering support and improve online-order experience for lower-performing restaurants

Expected Impact: Higher customer satisfaction, improved ratings, and increased order volume

Target premium, high-engagement segments

Insight: Table booking and premium segments show higher engagement and spend

Recommendation: Target premium/high-engagement segments with loyalty programs, reservations, and premium dining bundles

Expected Impact: Higher retention, greater revenue per customer, stronger premium segment growth

Impact & Value

Promote Mid-Tier Restaurants

Impact: 5% traffic redistribution, 5–8% higher utilization, ~8% engagement uplift

Value: Better revenue balance and marketplace efficiency

Expand Digital Ordering Support

Impact: ~10% higher order volume, ~5% better ratings

Value: Stronger retention and scalable digital growth

Target Premium Segments Strategically

Impact: ~7% higher spend, ~5% retention uplift

Value: Higher ROI from targeted promotions

Cost of Inaction

- Demand stays concentrated in a few restaurants
- Underperforming outlets remain underutilized
- Promotional spend stays inefficient



Projected Overall Impact

- 8–15% engagement uplift
- 7–10% revenue upside

Limitations, Data Gaps & Next Steps

Limitations

- Ratings concentration around 3.7 may affect inference
- Cross-sectional data only (no time-series forecasting)
- Limited business variables (no profit, delivery times, demographics)
- K-Means and Random Forest are exploratory, not production-grade

Additional Data Needed

- Customer-level behavioral data (repeat orders, basket size, retention)
- Time-based data (seasonality, peak hours, demand trends)
- Operational metrics (profit margins, delivery SLA, promotions)

Next Steps

- Build production-grade predictive model for ratings/demand forecasting
- Deploy real-time Tableau/BI dashboard with live monitoring
- Extend to recommendation engine using larger datasets

Current analysis provides strong directional insights; richer data can convert this into decision automation.